

SFT V12 Side Note — Baryogenesis Blender V8F

Derived Electroweak Solvency Forcing, Sphaleron Shutoff, and a Varying-Top-Yukawa Source Lane

Solvency Field Theory working note

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Abstract

This note records the V1–V8F baryogenesis-blender sequence. The question is not whether SFT derives CP violation or replaces the Standard Model. The narrower question is whether SFT supplies a derived electroweak solvency-forcing feature that can participate in baryogenesis when coupled to ordinary Sakharov-style particle-physics inputs.

The answer from the toy sequence is affirmative within the model limits. V5 found a raw electroweak solvency drop

$$\Delta\mathcal{U}_{\text{EW}} \simeq 0.174,$$

from a Standard-Model-like thermal inventory with numerical boson/fermion thermal integrals. V6 showed this result survives sphaleron-style washout gates. V7 inverted the problem and found that the required effective source scale is

$$\epsilon_{\text{eff}} \sim 10^{-8}.$$

V8A–V8F narrowed the viable source lane to a Higgs/top/electroweak-sector interface. An explicit varying-top-Yukawa source overlapping the restricted top+ $W/Z/H$ SFT forcing remains in the same source-strength band, robustly across electroweak gate choices.

The current best result is:

$\epsilon_{\text{source}} \sim 10^{-8}\text{--}10^{-7}$

for a varying-top-Yukawa source coupled to the top+ $W/Z/H$ electroweak solvency forcing under sphaleron-style washout. This is not a solution of baryogenesis; it is a Standard Model interface result. SFT supplies the derived electroweak solvency forcing. Particle physics must still supply the CP-violating source and baryon-number-violating microphysics.

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1 Scope and Claim

This note should be read as a Standard Model interface test. SFT takes the particle inventory, electroweak transition, CP-violating bias, and baryon-number-violating microphysics as external inputs. Its contribution is the boundary-accounting/solvency channel.

Claim: SFT ledger weighting of the electroweak thermal inventory generates a derived electroweak solvency-forcing feature. When coupled to a plausible Higgs/top source lane and sphaleron-style shutoff, this forcing can fossilize a baryon asymmetry of the observed scale in the toy model.

This is not the claim:

SFT derives CP violation from first principles.

Nor:

SFT fully solves baryogenesis.

The correct framing is:

Baryogenesis is a solvency-biased electroweak phase-transition fossil.

The narrowed V8F form is:

A varying-top-Yukawa source couples to the SFT top+W/Z/H electroweak solvency forcing.

2 Physical Picture

The electroweak transition is the natural baryogenesis interface for SFT because it is where the Higgs vacuum expectation value turns on. In SFT language:

Higgs VEV turns on $\rightarrow$ rest-mass clocks turn on $\rightarrow$ ledger burden reorganizes $\rightarrow$ solvency forcing appears $\rightarrow$ bary

The frozen remnant is

$$Y_B = \frac{n_B - n_{\bar{B}}}{s},$$

where s is entropy density. The reference scale used throughout is

$$Y_{\text{ref}} \simeq 8.5 \times 10^{-11},$$

corresponding to the usual baryon-to-photon scale $\eta_B \sim 6 \times 10^{-10}$, up to the standard entropy/photon conversion factor.

3 Baryogenesis Engine

The core toy equation is

$$\frac{dY_B}{dx} = \epsilon_{\text{CP}} \left(\frac{\Gamma_B}{H} \right) F_{\text{solvency}}(T) - \left(\frac{W}{H} \right) Y_B, \quad x = \ln \left(\frac{T_{\text{start}}}{T} \right).$$

Here ϵ_{CP} is the CP-violating bias, Γ_B/H is the baryon-number-violating source strength relative to expansion, W/H is the washout rate relative to expansion, and $F_{\text{solvency}}(T)$ is the SFT-derived forcing.

The electroweak order parameter is modeled by

$$\Phi(T) = \frac{1}{2} \left[1 + \tanh \left(\frac{\ln(T_c/T)}{\Delta} \right) \right], \quad q(T) = \frac{v(T)}{v_0} = \sqrt{\Phi(T)}.$$

The transition activity is

$$\frac{d\Phi}{dx} = \frac{2\Phi(1-\Phi)}{\Delta}.$$

The earliest toy washout was

$$\frac{W}{H} = K_0(1-\Phi)^p,$$

while the V6+ sphaleron-style gate is

$$\boxed{\frac{W_{\text{sph}}}{H} = K_0 \exp \left[-\alpha \frac{v(T)}{T} \right].}$$

4 SFT Ledger Weighting

At temperature T , each species contributes an energy density

$$\rho_i(T) = \frac{g_i T^4}{2\pi^2} \int_0^\infty \frac{p^2 \sqrt{p^2 + (m_i/T)^2}}{\exp \left(\sqrt{p^2 + (m_i/T)^2} \right) \pm 1} dp,$$

with the plus sign for fermions and the minus sign for bosons. The SFT ledger-weighted inventory is

$$\rho_{\text{ledger}}(T) = \sum_i w_i(T) \rho_i(T), \quad w_i(T) = 1 + R_i(T),$$

where the relativistic fraction is approximated by

$$R_i(T) = \frac{1}{1 + \left(\frac{m_i(T)/T}{y_0} \right)^n}.$$

Thus radiation-like species pay roughly $2\times$, and massive/ticking-like species pay roughly $1\times$. Higgs-coupled species have

$$m_i(T) = m_{i,0} q(T).$$

The solvency estimate used in V5–V8 is

$$\boxed{\mathcal{U}(T) = \frac{2\pi}{3} \frac{\rho_{\text{ledger}}(T)}{\rho_{\text{energy}}(T)}}.$$

This comes from $\mathcal{U} = \dot{S}/(HI)$ using a Hubble volume, horizon information capacity, complete-cycle operation counting, and the SFT closure $H^2 = \rho/(4\pi^2)$.

5 Candidate SFT Forcings

Three raw, unnormalized forcing candidates were tested:

$$F_{d\mathcal{U}}(T) = \max \left(0, -\frac{d\mathcal{U}}{dx} \right), \tag{1}$$

$$F_{d\ln\mathcal{U}}(T) = \max \left(0, -\frac{d\ln\mathcal{U}}{dx} \right), \tag{2}$$

$$F_{\text{mass}}(T) = \left| \frac{dq}{dx} \right| \frac{\rho_{\text{Higgs-coupled}}}{\rho_{\text{total}}} \Delta\mathcal{U}_{\text{EW}}. \tag{3}$$

Numerically, $F_{d\ln\mathcal{U}}$ is the best all-around fractional forcing. Conceptually, F_{mass} is the cleanest Higgs-clock turn-on object. V8E/V8F then introduced a restricted top+ $W/Z/H$ forcing, F_{t+WZH} , to isolate the electroweak/Higgs-contact sector.

6 V1–V5: Removing Free Choices

The V1–V5 sequence progressively replaced free assumptions with physical structure:

1. **V1:** arbitrary source/washout windows; proof of concept.
2. **V2:** transition order parameter Φ with source proportional to $|d\Phi/d\tau|$.
3. **V3:** temperature-anchored transition using $x = \ln(T_{\text{start}}/T)$.
4. **V4:** derived electroweak forcing shape from a Standard-Model-like thermal inventory plus SFT ledger weights.
5. **V5:** numerical boson/fermion thermal integrals and raw $\mathcal{U}(T)$ -derived amplitudes.

For the reference electroweak parameters

$$T_c = 160 \text{ GeV}, \quad \Delta = 0.22,$$

V5 found

$$\mathcal{U}_{\text{pre}} = 4.18879, \quad \mathcal{U}_{\text{post}} = 4.01448, \quad \boxed{\Delta\mathcal{U}_{\text{EW}} \simeq 0.174.}$$

Table 1: V5 reference results.

Forcing	Final Y_B	Ratio to reference	A_{required}
$F_{d\mathcal{U}}$	2.65×10^{-10}	3.11	0.321
$F_{d\ln\mathcal{U}}$	6.46×10^{-11}	0.759	1.32
F_{mass}	4.69×10^{-11}	0.552	1.81

All three are order-unity. The fractional forcing $F_{d\ln\mathcal{U}}$ is closest by default.

7 V6: Sphaleron-Style Washout

V6 replaced the toy washout gate with sphaleron-style gates tied to $v(T)/T$. The best physical gate was

$$\frac{W}{H} = K_0 \exp[-\alpha v(T)/T].$$

For this gate and $F_{d\ln\mathcal{U}}$, V6 tested $N = 728$ runs and found:

- 63.7% within a factor of 10 of Y_{ref} .
- 29.7% within a factor of 3.
- 33.5% with order-unity required amplitude.
- Median $A_{\text{required}} \simeq 1.36$.
- Median freeze temperature $T_{\text{freeze}} \simeq 97.2 \text{ GeV}$.

So the V5 result survives sphaleron-style washout.

8 V7: Source Normalization as a Number

V7 inverted the source problem. Since the toy equation is linear in source strength,

$$Y_B \propto \epsilon_{\text{eff}}, \quad \epsilon_{\text{eff}} = \epsilon_{\text{CP}} G_{\hat{B}} A.$$

Using V6 runs with $\epsilon_{\text{CP}} = 10^{-8}$, we infer

$$\epsilon_{\text{eff,required}} = 10^{-8} A_{\text{required}}.$$

For the best physical architecture, V7 found

$$\epsilon_{\text{eff,required,median}} \simeq 1.36 \times 10^{-8}.$$

This is the first hard boundary. SFT does not derive CP violation here; it says the particle-physics source must supply an effective CP/source strength near 10^{-8} .

9 V8A: Source Feasibility Map

V8A asked whether $\epsilon_{\text{eff}} \sim 10^{-8}$ is plausible for electroweak baryogenesis source classes. The conclusion was:

- Minimal CKM-only electroweak baryogenesis is not the right lane.
- Higgs-sector, varying-Yukawa, dynamical-CKM, composite-Higgs, or dark-sector CP sources remain plausible lanes.
- The best conceptual match is varying-Yukawa/dynamical-CKM, because the source is active while the Higgs VEV and rest-clock structure turn on.

The SFT bridge is

$$m_i(T) = \frac{y_i(T)v(T)}{\sqrt{2}}, \quad \nu_{C,i}(T) = \frac{m_i(T)c^2}{h},$$

so

$$\frac{d \ln \nu_{C,i}}{dx} = \frac{d \ln y_i}{dx} + \frac{d \ln v}{dx}.$$

This links V11.3 Higgs contact, varying Yukawa sources, and electroweak solvency forcing.

10 V8B: Species-Decomposed Higgs-Contact Drop

V8B decomposed $\Delta\mathcal{U}_{\text{EW}}$ by species. The total was

$$\mathcal{U}_{\text{pre}} = 4.18879, \quad \mathcal{U}_{\text{post}} = 4.01457, \quad \Delta\mathcal{U}_{\text{EW}} = 0.17423.$$

The key finding is that the electroweak solvency feature is concentrated in the Higgs-contact sector:

$$t, W, Z, H.$$

Table 2: V8B grouped leave-one-out contribution to $\Delta\mathcal{U}$.

Group	Leave-one-out $\Delta\mathcal{U}$ contribution
Top + $W/Z/H$	0.159
Top only	0.0904
$W/Z/H$ only	0.0690

Thus the SFT baryogenesis forcing is not a generic thermal-inventory artifact. It is genuinely an electroweak/Higgs-contact feature.

11 V8C: Source-Window Overlap

V8C modeled a generic Higgs/top/Yukawa-like CP source window,

$$S_{\text{CP}}(T) \sim \left| \frac{dy_t}{dx} \right|,$$

and tested overlap with $F_{d\ln\mathcal{U}}$ under sphaleron washout. Source windows centered near or slightly below the electroweak transition work best:

$$T_{\text{source}} \sim 80\text{--}160 \text{ GeV},$$

with broad viable windows across roughly 100–200 GeV. Early/hot windows are washed out; broad or slightly late source windows survive better. The source requirement remained near 10^{-8} .

12 V8D: Explicit Varying-Top-Yukawa Source

V8D replaced the generic source with an explicit varying-top-Yukawa toy:

$$y_t(T) = y_{t,0} [1 + \delta_y Y(T)], \quad S_y(T) = \left| \frac{d \ln y_t}{dx} \right|.$$

The engine was

$$\frac{dY_B}{dx} = \epsilon_{\text{source}} S_y(T) F_{d\ln\mathcal{U}}(T) - \frac{W}{H} Y_B.$$

For $K_0 = 1000$, V8D found:

Table 3: V8D explicit varying-top-Yukawa source results.

α	δ_y	Median required ϵ_{source}
2.0	0.5	8.71×10^{-8}
2.5	0.5	2.54×10^{-8}
3.0	0.5	1.55×10^{-8}
2.5	1.0	1.60×10^{-8}
3.0	1.0	9.20×10^{-9}
3.0	2.0	5.95×10^{-9}

Thus an explicit varying-top-Yukawa source preserves the target

$$\epsilon_{\text{source}} \sim 10^{-8}\text{--}10^{-7}.$$

13 V8E: Electroweak-Restricted Top-Sector Forcing

V8E asked whether late, low-temperature threshold structure in the full $F_{d\ln\mathcal{U}}$ was carrying the result. It restricted the forcing to the top+ $W/Z/H$ sector:

$$F_{t+WZH}.$$

For $K_0 = 1000$, $\alpha = 2.5$, V8E found:

Table 4: V8E restricted top+ $W/Z/H$ forcing results.

δ_y	Median $\epsilon_{\text{source,required}}$ for F_{t+WZH}
0.10	1.14×10^{-7}
0.20	5.98×10^{-8}
0.50	2.79×10^{-8}
1.00	1.74×10^{-8}
1.50	1.40×10^{-8}
2.00	1.22×10^{-8}

For stronger sphaleron shutoff, $\alpha = 3.0$, the same restricted forcing gave:

Table 5: V8E restricted forcing with stronger sphaleron shutoff.

δ_y	Median $\epsilon_{\text{source,required}}$
0.50	1.71×10^{-8}
1.00	1.01×10^{-8}
1.50	7.82×10^{-9}
2.00	6.68×10^{-9}

This removed the contamination concern. The top/electroweak/Higgs sector alone can carry the result.

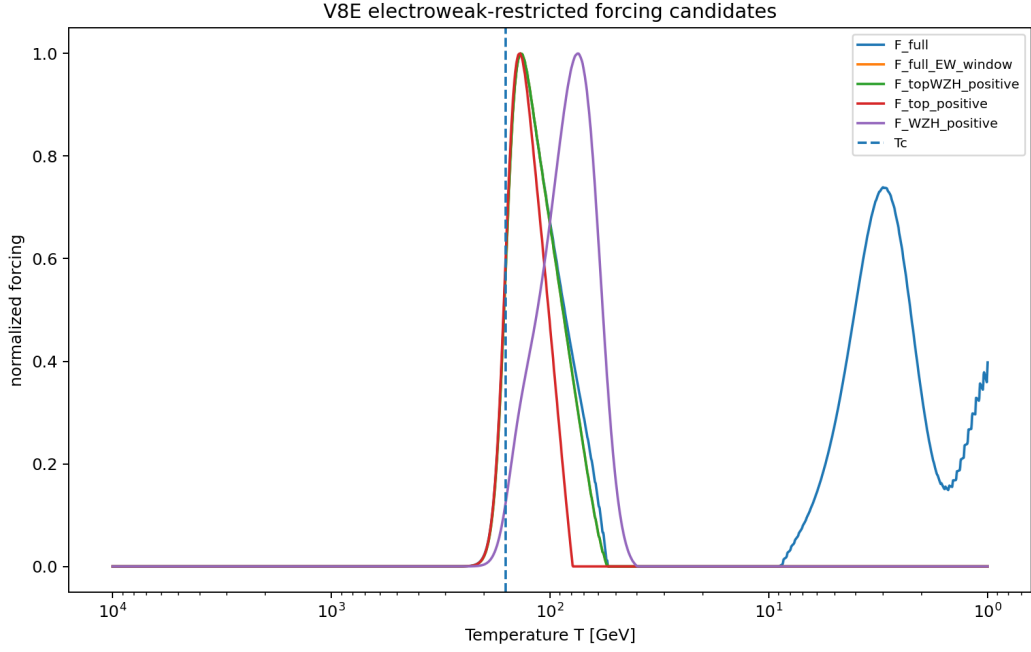


Figure 1: V8E electroweak-restricted forcing profiles.

14 V8F: Robustness to Electroweak Gate Choice

V8F tested whether the restricted top-sector result depended on the chosen electroweak gate window. The gate boundaries were varied across:

$$T_{\text{low}} = 50, 60, 80, 100 \text{ GeV}, \quad T_{\text{high}} = 220, 260, 320, 400 \text{ GeV},$$

with multiple edge sharpness values.

The result passed. For F_{t+WZH} , the median required source across gate choices was:

Table 6: V8F robustness to electroweak gate choice.

α	δ_y	Median ϵ_{source} across gates
2.5	0.5	3.09×10^{-8}
2.5	1.0	1.96×10^{-8}
2.5	2.0	1.39×10^{-8}
3.0	0.5	1.75×10^{-8}
3.0	1.0	1.10×10^{-8}
3.0	2.0	7.72×10^{-9}

Thus the V8E result is not an artifact of a single electroweak gate choice.

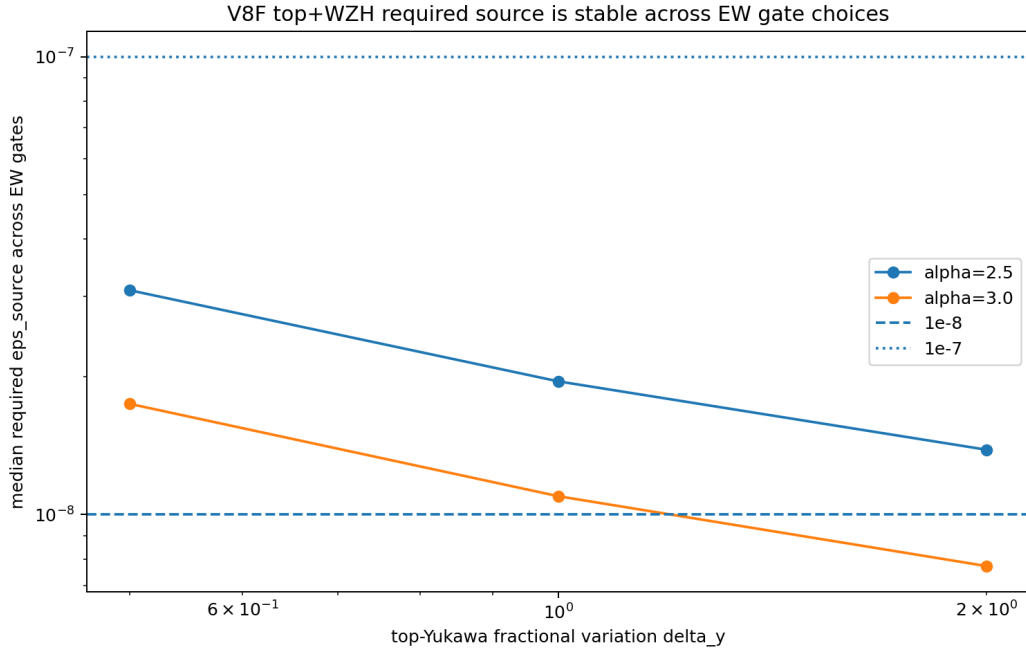


Figure 2: V8F: required source strength remains in the 10^{-8} band across electroweak gate choices.

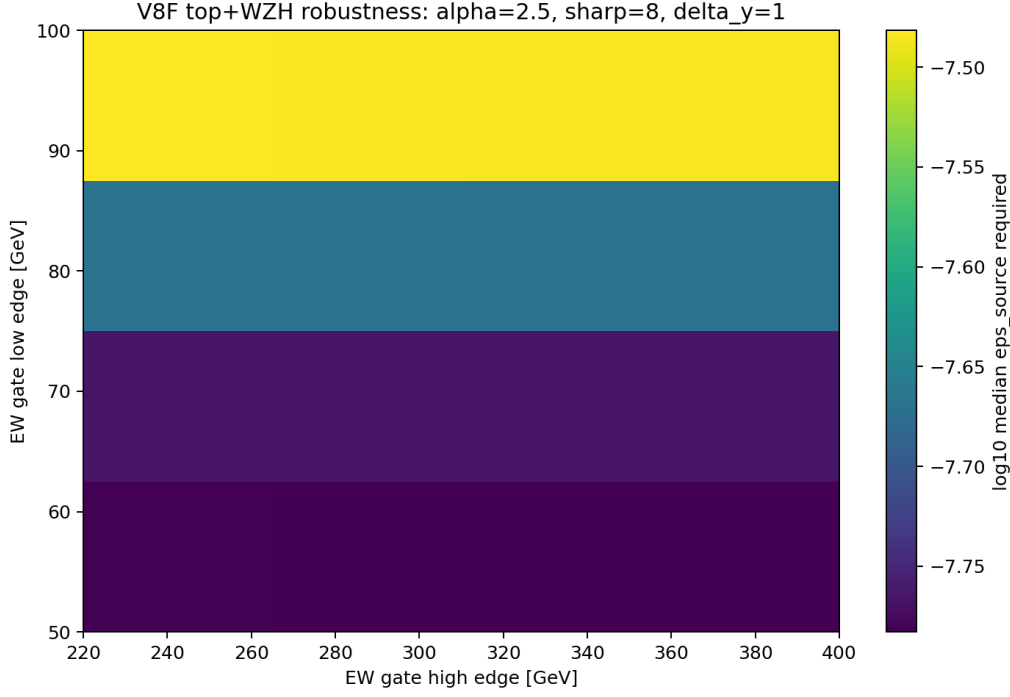


Figure 3: V8F heatmap for the top+W/Z/H gate robustness test.

15 Current Best Narrowed Claim

The current claim is no longer broad baryogenesis. It is specific:

A varying-top-Yukawa source overlapping the SFT top/electroweak/Higgs-contact forcing can produce the observed baryon-fossil scale with an effective source coefficient of order 10^{-8} .

The guardrails are:

1. SFT supplies the derived electroweak solvency forcing.
2. The particle-physics sector must still supply CP violation and baryon-number-violating source physics.
3. Minimal CKM-only electroweak baryogenesis is not claimed.
4. The result is a toy Standard Model interface result, not a completed baryogenesis theory.

16 What This Does Not Establish

This note does not establish:

1. A first-principles derivation of CP violation.
2. A first-principles derivation of baryon-number violation.
3. A complete electroweak baryogenesis model.
4. A precision finite-temperature Standard Model calculation.
5. A proof that the Standard Model alone has enough CP violation.
6. A replacement for existing baryogenesis mechanisms.
7. A unique prediction of Y_B without particle-physics inputs.
8. A proof that varying-top-Yukawa electroweak baryogenesis is realized in nature.

The SFT contribution is the derived electroweak solvency forcing and its demonstrated overlap with a plausible Higgs/top source lane.

17 Open Problems

The narrowed open problems are:

1. **Concrete source model:** replace the toy varying-top-Yukawa source with a concrete electroweak baryogenesis source model.
2. **Real sphaleron normalization:** replace $W/H = K_0 \exp[-\alpha v(T)/T]$ with a literature-calibrated or lattice-inspired sphaleron washout rate.
3. **Finite-temperature Higgs profile:** replace the tanh Higgs crossover with a more realistic finite-temperature electroweak profile.
4. **SFT forcing derivation:** derive whether the correct forcing object is $F_{d\mathcal{U}}$, $F_{d\ln\mathcal{U}}$, F_{mass} , F_{t+WZH} , or a combination.
5. **Ledger-weight derivation:** derive $w_i(T) = 1 + R_i(T)$ from SFT operation counting rather than choosing it as a thermal bridge.

18 Earned Statement

A Standard-Model-like electroweak thermal inventory with SFT ledger weights generates a derived electroweak solvency-forcing feature. This feature has the right raw scale to participate in baryogenesis, survives sphaleron-style washout, and remains viable when restricted to the top plus electroweak/Higgs sector. When coupled to an explicit varying-top-Yukawa source window, the top+ $W/Z/H$ SFT forcing produces the observed baryon-fossil scale for effective source coefficients in the 10^{-8} to 10^{-7} band, robustly across electroweak gate choices. The CP-violating source and baryon-number-violating microphysics remain particle-physics inputs; SFT's contribution is the derived electroweak solvency forcing channel.

Compact form:

Baryogenesis is a solvency-biased electroweak phase-transition fossil.

Specific V8F form:

The viable interface is a varying-top-Yukawa source coupled to the SFT top+ $W/Z/H$ electroweak solvency forcing.

One-Line Summary

The electroweak transition turns on Higgs-generated rest clocks. SFT ledger weighting says that this reorganizes the universe's solvency burden. The top+ $W/Z/H$ sector carries the relevant electroweak forcing, and a varying-top-Yukawa source can overlap that forcing while sphaleron washout shuts off, leaving a baryon fossil in the observed range.